

nastran95ase Theoretical Manual

What the numbers in a `nastran95ase` print file and its OUTPUT4 export mean: which NASA rigid formats run, how the doublet-lattice aerodynamics, the generalized matrices, the splines, the PK flutter solution, the modes and the gust response are computed in this fork, what the fork changed in them, and what the results have been checked against. It is compiled from the fork's `HALO.md`, the sources in `mis/`, `msc/`, `rf/` and `bin/nastrn.f.in`, the VehicleDesign skills `nastran-analysis` (with its distillation of NASA's User's Manual Volume II, `references/users-manual-vol2-flutter-gust-modes.md`, cited below as "User's Manual Vol. II"), `nastran-flutter`, `nastran95-performance` and `build-nastran95`. Every mechanism names the routine that does it. Where the sources are silent the text says "not documented". The companion `nastran95ase_users_manual.md` covers running decks.

1. The rigid formats used

NASTRAN-95 has no solution sequences in MSC's sense; it has rigid formats, DMAP programs in `rf/` selected by `APP` and `SOL n,0`. The front end's map (`msc/mscexec.c`):

MSC	rigid format	what the fork uses it for
SOL 101, 105, 107-112	DISPLACEMENT 1, 5, 7-12	statics, buckling, complex eigenvalues, frequency and transient response
SOL 103	DISPLACEMENT 3 (<code>rf/DISP03</code>)	normal modes; the modes run of every flutter restart; the <code>.phg / .mzg</code> export
SOL 145	AERO 10 (<code>rf/AERO10</code>)	modal flutter
SOL 146	AERO 11 (<code>rf/AERO11</code>)	modal aeroelastic response (gust)

Both aero formats begin with the whole of rigid format 3 in line: GP1 .. GP4, MCE/SCE/SMP, RBMG1-4 (the SUPORT rigid-body partition), DPD, READ (the real modes), MTRXIN (`K2PP / M2PP / B2PP` direct input), GKAD, GKAM (modal matrices), APD (the aerodynamic model), GI (the spline), AMG (the aerodynamic influence matrices), AMP (the generalized aerodynamics); then AERO 10 runs the flutter loop FA1 -> (CEAD, K method only) -> VDR -> FA2 -> REPT LOOPTOP,100, the V-g plots, MODACC, ADR (aerodynamic loads) and the physical recovery DDR1/SDR1/MPYAD/UMERGE/SDR2; AERO 11 runs FRLG, GUST, FRRD2, IFT, MODACC, ADR, VDR/SDR1-3, XYTRAN/XYPLOT and RANDOM (User's Manual Vol. II §3, §4). COSMIC has no file-based modal transfer: the only cross-run carry is checkpoint/restart (section 11), and the fork's OUTPUT4 alter is the only route for matrices to ZAERO or to the state-space plant (section 10).

DIAG 14 prints the numbered DMAP listing (how an ALTER point is re-found), DIAG 16 the FEER diagnostics, DIAG 39 traces the PK iteration from inside FA1PKA.

2. The structural model as the solver receives it

The translation (`msc/mscxlat.c`) changes no numbers but it changes the model's description, and the print reflects that:

- Rigid bars and RBE2s are CRIGD1 constraints (the same constraint; COSMIC wants the independent grid first, chosen so a SUPORT ed or SPC ed grid is never dependent).
- A coincident CBUSH is six CELAS2 springs, one per non-zero stiffness; a separated one is a CONROD with K1 only, with a unit-modulus MAT1 made for it.

- Degrees of freedom nothing is attached to are constrained (what MSC's AUTOSPC does); NASTRAN-95 would otherwise factor a singular stiffness and stop with UFM 3097.
- Ids above $2^{24} - 1$ are renumbered everywhere, `PARAM,GRDPNT` included, and put back in the print.
- `WTMASS`, `COUPMASS`, `GRDPNT` are applied as in MSC; `WTMASS` in EMA.
- A `SUPPORT` (from `SUPPORT1`) sets `REACT:RBMG1-4 build [D] = -[K11]^-1[K1r]`, the rigid-body error ratio and `[mr]`, and `READ` constructs the rigid-body modes rather than extracting them (User's Manual Vol. II §2). The rigid-body mass matrix on the `SUPPORT` freedoms must be non-singular (UFM 2200): with `CONM2`s carrying no rotary inertia, a rotation about an axis all the masses lie on has none.

The mass matrix of every repo model is only semi-definite - lumped `CONM2`s with no `I11`, `I22`, `I33` leave the rotations massless - and that is the one property of the model the eigensolver was not written for (section 3.2).

3. Normal modes

3.1 FEER

`EIGRL` becomes `EIGR FEER` with the shift at `V1`, or 0.5 Hz when `V1` is blank (`do_eigr1`, `msc/mscxlat.c`): `FEER` ("Fast Eigenvalue Extraction Routine") is a tridiagonal reduction (`mis/ferxtd.f`, `mis/ferxts.f`) followed by a QR iteration on the reduced matrix (`mis/fqrwv.f`, `mis/fqrw.f`). It works in the mass metric: trial vectors are mass-orthogonalised and the norm that goes under the square root at label 480 of `FERXTD` is the mass norm of the new trial vector. A zero shift on a free-free model whose rigid-body modes are only nearly zero loses every mode; the manual's "small positive internal shift" and up to three shifts exist to remove stiffness singularities, and the user shift is given in Hz and converted to radians squared (User's Manual Vol. II §2; skill `build-nastran95`). `FEER` returns more roots than asked (a reduced problem of twice the size); the print rewrite and the `.phg` writer cut to the number requested. Modes are numbered by algebraic eigenvalue, and with `MASS` normalisation the generalized stiffness equals the eigenvalue, which is what `read_khh` reads off the `REAL EIGENVALUES` table. The summary `OEIGS` gives the reason for termination: 0 normal, 1 fewer than requested, 3 the problem size was reduced; a mode whose theoretical error bound exceeds the `EIGR` allowable (default $0.001/n$) is marked `FAIL` and dropped - a second way to come back short of the count. The `TIME` card is an eigensolver input in NASA's code: an undersized `TIME` truncates the mode count with a clean `END OF JOB` (User's Manual Vol. II §2); the translated decks carry a large `TIME` and the wall-clock watchdog instead.

3.2 The reseeding and "fewer modes" (the fork's guards)

On a semi-definite mass matrix roundoff put the mass norm at $-1e-17$ on row 88 of 250 of an 830-grid deck, `DSQRT` returned NaN, every test after it was a comparison a NaN fails, and rows 88 to 250 came out NaN: the solver spun for hours. The fork (`HALO.md`, "Never hang"):

- treats a norm that is not positive (or is NaN) as the null trial vector it is: the reduction reseeds through its existing restart loop (`FEER3`, capped at `MORD` reseeds), and `mis/fernpd.f` prints UWM 2394 once (`FEER TRIAL VECTOR n HAS NO POSITIVE MASS NORM ... THE VECTOR IS DISCARDED AND THE REDUCTION RESEEDDED; IF FEWER MODES THAN REQUESTED FOLLOW, SEE MESSAGE 2390`). The same guard is on the two start-vector normalisations (labels 40 and 240) and a NaN test joins the "problem size reduced" exits (150, 310);
- writes the in-memory trial vectors (NASA's 1994 `NIDORV` in-core mode) to scratch file 7 before every return to the reseed loop, because the reseed read them from there and found nothing (`I/O SUBSYSTEM ERROR NUMBER 110 ... SCRATCH7 after END OF JOB`, exit 0, no modes). On a modern open core the vectors always fit, so the reseed path had never worked in this build;

- bounds the QR sweep (FQRWV / FQRW label 70) at 200 sweeps per eigenvalue plus 1,000, or a NaN in its input, with UFM 2395 and MESSAGE -37.

What the model then gets: after a reseed on this metric the next vector's orthogonalisation converges by a factor of 0.6 a pass and each vector after it slower still; raising NASA's cap of 14 passes to 60 bought four rows and nothing else, so the cap stays and the shortfall is reported (UWM 2390 gives the count). The 830-grid deck returns 89 accurate modes of 120 in 7 seconds; MSC's Lanczos is not affected by the metric. The translator counts the CONM2s without rotary inertia and warns up front (UWM 9133) with the cure: rotary inertia on the lumped masses, or fewer modes.

3.3 Mass normalisation and the exported matrices

With MASS normalisation $\phi^T M \phi$ is the identity, which is the check VehicleDesign's four-way test applied to the .phg and .mzg export against the weight generator (HALO.md, "What the front end is checked against"): a mass matrix with one spurious entry per empty degree of freedom still factors, still gives modes, and still looks like a mass matrix - which is why null columns are left out of the OUTPUT4 file (section 10).

4. The modal matrices (GKAM)

GKAM USETD,PHIA,,LAMA,DIT,M2DD,B2DD,K2DD,CASECC/MHH,BHH,KHH,PHIDH/NOUE/C,Y,LMODES=0/C,Y,LFREQ=0./C,Y,HFREQ=-1.0/.../C,Y,KDAMP (rf/AERO10 line 422) forms the generalized mass, damping and stiffness and the modal matrix PHIDH on the d-set (User's Manual Vol. II §3):

- $[M_{hh}] = \text{diag}(m_i) + \phi^T M_{2dd} \phi$, $[K_{hh}]$, $[B_{hh}]$ likewise with the K2PP / B2PP direct input matrices the case control names (kept by the front end since 2026-09-26; the trim-load study's per-subcase steady-lift K2PP / B2PP DMIGs reach the children).
- The modes kept are PARAM LMODES (the n lowest) or LFREQ / HFREQ (a window in Hz, with LMODES=0); the DMAP defaults select nothing, so one of them is required. The EIGR count only sets how many modes READ finds.
- Damping from TABDMP1 (SDAMPING): with KDAMP = -1 (NASTRAN-95's default) $b_i = m_i 2 \pi f_i g(f_i)$, viscous equivalents, in $[B_{hh}]$; with KDAMP = +1 $k_i = (1 + i g(f_i)) (2 \pi f_i)^2 m_i$, complex stiffness and no B (the fork's GKAM1A does exactly that). MSC and Simcenter number it the other way round (their default +1 is viscous), so without the parameter both codes are viscous and the same, and an explicit PARAM,KDAMP means the opposite in the two; the front end passes it unchanged and no repository deck sets it.

5. The aerodynamics

5.1 APD: the aerodynamic model and the (Mach, k) list

APD builds the aerodynamic model from the EDT bulk data (CAERO1/PAERO1 boxes, AERO, MKAERO1/MKAERO2, SPLINE1/SPLINE2, SET1, FLFACT, FLUTTER) and emits the SPLINE, AERO, ACPT, FLIST blocks, NK aero degrees of freedom, NJ boxes and BOV = b/V from the AERO card's velocity field (User's Manual Vol. II §3; HALO.md quotes 4,564 k-set rows against the monarch's 2,282 boxes). The (Mach, k) pairs of the MKAERO1 and MKAERO2 cards are collected, sorted by Mach then k (APDMKS, a stable insertion sort at the end of mis/apd.f) and written to the AERO data block once, and the list is printed as UIM 9457. NASA wrote the pairs card by card in sorted bulk data order, and MKAERO1 cards of one Mach set tie on every sorted field, so a k list split over several cards came out with the cards shuffled - the same in Simcenter, which does not sort. The flutter and gust interpolations never cared (they interpolate over the

pairs of a Mach wherever they sit); the order matters only to a reader of the exported QHHL / QHJL / QKHL blocks, which follow this list. PK roots are unchanged to round-off by the sort (HALO.md, "The order of the (Mach, k) pairs").

The reduced frequency throughout is $k = \omega b / V$ with $b = \text{REFC} / 2$, the semichord: FA1 halves REFC itself, BREF in FA1PKE is b, and BOV = BREF/VEL (mis/fa1pke.f; skill nastran-flutter §8).

5.2 GI: the spline

GI forms [Gka], the spline from the structural a-set to the aerodynamic k-set, from the SPLINE1 (surface) and SPLINE2 (beam) cards and their SET1 lists; GTKA is its d-set form used by AMP (rf/AERO10 line 546) and MPYAD GTKA,CPHIA,/CPHIK puts a structural eigenvector through it onto the boxes for the COMPLEX EIGENVECTOR tables (plunge in T3, pitch in R2, in the box's frame). The spline formulas themselves are in NASA's Theoretical Manual, not in the sources here: not documented. The front end strips MSC's METH and USAGE fields from SPLINE1.

5.3 AMG: the doublet lattice

AMG AERO,ACPT/AJL,SKJ,D1JK,D2JK/NK/NJ (rf/AERO10 line 516) computes, for every (Mach, k) pair, the aerodynamic influence matrix AJJ (boxes by boxes, complex), the integration (area) matrix SKJ and the downwash matrices D1JK (the displacement part) and D2JK (the velocity part, multiplied by ik in AMP). The lattice is built a row (receiving box) at a time by GEND: for each receiving box DPPS sums the kernel of every sending box (SUBP -> SNPDF the steady part, INCRO -> TKER the incremental oscillatory kernel, IDF1 / IDF2 its integrals) (mis/gendp.f header).

NASA's parabolic kernel (1971). INCRO fits the numerators of the planar (P1) and nonplanar (P2) incremental kernels across the sending box's doublet line by a parabola through three points, and TKER approximates the kernel integrals I1, I2 with Laschka's 11 exponential terms. One arctangent is in the wrong quadrant: IDF1 / IDF2 take $\text{ATAN}(2e|\text{zeta}|/(r^2 - e^2))$ (Rodden 1971) without Rodden's 1972 correction, so for a receiving point off the sending box's plane ($|\text{zeta}|/e > 0.001$) and within $r < e$ of its centre the angle belongs in the second quadrant and NASA's element is off by order one at 199 of 3,000 random geometries. A tail box lined up spanwise with a wing box a little above or below it is that case. It is left as NASA wrote it, so the parabolic path stays NASA's to the bit (HALO.md, "The quartic doublet-lattice kernel").

The quartic kernel (NASTRAN SYSTEM(270)=1, mis/dlmq.f). What MSC and Simcenter call QUARTICDLM, "the new quartic formulation of the doublet lattice kernel (N5KQ)": Rodden, Taylor and McIntosh, "Further Refinement of the Subsonic Doublet-Lattice Method", J. Aircraft 35(5), 720-727, 1998. The numerators of P1 and P2 are fitted by a quartic through five points ($\eta = -e, -e/2, 0, e/2, e$), the closed-form integrals extended to match (the paper's eqs. 15-19, 28-32 for the fit, 20-25, 33-34 for the integrals), and I1, I2 approximated with Desmarais' 12 exponential terms. Both parts are needed to reproduce Simcenter. INCROQ is INCRO for a batch of reduced frequencies, TKERQ is TKER for a batch with the series chosen (Laschka or Desmarais), DLMQFT the fit, DLMQNP the nonplanar integral; double precision, no COMMON, no SAVE, the geometry, regime, logarithm and arctangent once per element for the whole batch, the arctangent in its quadrant. The closed forms are those DLR's PanelAero writes (BSD-3), checked against mpmath quadrature: exact to $1e-13$ where the paper's forms are exact and $1e-4$ where its series stands in. One change from PanelAero: in the coplanar limit ($|\text{zeta}|/e \leq 0.001$) the nonplanar term takes the series alpha of eq. 25, which the quadrature bears out; PanelAero's value there, $2e/(\eta^2 - e^2)$, does not, and NASA's IDF2 uses a third. It matters little, that numerator carrying zeta and being near zero. With the Laschka series TKERQ reproduces NASA's TKER to its single precision. INCRO and AMGK's INCROK hand over to INCROQ when N95DLQ() is 1 (msc/mscisa.c); the aerodynamic cache keys carry the kernel (IAECV 1 parabolic, 3 quartic). On the benchmark HALE wing the parabolic kernel alone put the fork's

flutter 3 % below Simcenter's (30.62 against 31.57 m/s). Cost on the five-Mach monarch deck: AMG 110 CPU s parabolic, 615 quartic (five kernel evaluations in double precision against three in single).

How the fork computes it faster without moving a bit (HALO.md, "Linux" and "The second pass"; skill nastran95-performance §4): the rows are computed a block at a time on OpenMP threads and packed in GEND's order (mis/gendp.f; /DLM/ and /KDS/ THREADPRIVATE in TKER, INCRO, FLLD, SUBB); up to 8 pairs of one Mach are done in a batch with the steady part, the geometry, TKER's branches, square roots and exponential and IDF1/IDF2's logarithm and arctangent computed once and every k-dependent statement a loop over the batch (mis/amgk.f, mis/tkerv.f: TKERV, INCROK, SUBPK, IDF1V, IDF2V, DPPSK); the kernel routines are compiled -O2 -fautomatic -ffp-contract=off, with !GCC\$ NOVECTOR on the loops that call SIN, COS and EXP because gfortran would otherwise call glibc's libmvec, which rounds differently (194,796 of 200,000 test geometries differed). AJJ dumped as AMP reads it is byte for byte the serial NASA loop's.

5.4 AMP: the generalized aerodynamic matrices

AMP AJJL,SKJ,D1JK,D2JK,GTKA,PHIDH,D1JE,D2JE,USED,AERO/QHHL,QKHL,QHJL/NOUE/S,N,XQHHL/V,Y,GUSTAERO=-1 (rf/AERO10 line 546). Per (Mach, k) pair, as mis/ampk.f's header transcribes the serial path:

1. GKI = GTKA' PHIDH - the modes on the k-set (structure-dependent).
2. DJH = D1JK GKI + i k D2JK GKI - the modal downwash (SSG2C's two multiply-adds with multipliers (1, o) and (o, k)).
3. QJH = (AJJ')⁻¹ DJH - the modal pressures: AMPC transposes AJJ (TRANP1), decomposes the transpose out of core (CFACR) and solves the NOH right-hand sides (CFBSOR) in NASA's path; the fork's mis/ampcz.f unpacks AJJ transposed into open core as complex double precision, factors it in place with partial pivoting (the fork's LU, AMPCZB, or LAPACK's ZGETRF/ZGETRS through OpenBLAS, mis/ampczs/) and solves the right-hand sides, falling back to NASA's path when there is no core, a pivot vanishes or N95_INCORE_AJJ=0.
4. QKH = SKJ QJH - the box forces per mode (MMA214's "A not transposed", summed over SKJ's columns in order).
5. QHH = GKI' QKH (QIH in the code) - the generalized aerodynamic force matrix (MMA104's sums in their order).

The pairs' blocks are written side by side into QHHL (modes by modes per pair), QKHL (k-set by modes per pair) and, with GUSTAERO = +1, QHJL (the gust columns, section 9). In the expanded form of the task statement, QHHL = PHIDH' GTKA' SKJ (AJJ')⁻¹ (D1JK + i k D2JK) GTKA PHIDH per pair, per unit dynamic pressure: Q multiplies the modal displacements to give the modal aerodynamic force per unit dynamic pressure (skill nastran-flutter §8). The transpose on AJJ is AMPC's; whether NASA's AJJ is the pressure-to-downwash or the downwash-to-pressure relation is not documented in the sources beyond that.

The fork's AMP changes (HALO.md): the pair loop is a pipeline of OpenMP tasks - a read task per pair (its AJJ and SKJ off the files, kept open and read in order), a compute task (DJH, the solve, QKH, QIH) and a write task (QJHL, QHHL), every GINO call chained on one dependence in pair order, up to 16 pairs in core (N95_AMP_SLOTS, N95_AMP_MB), each solve on one thread (ZGETRF gives the same bits on 1 thread as on 32); AJJL and SKJ are no longer rewound and walked k-squared (mis/ampc.f, mis/ampd.f); the serial path skips terms with an unstored (zero) factor and the pipeline keeps them, and a zero of either sign added to a sum that started at +o changes nothing. The OpenBLAS solve is the one change that is not bit-identical: a blocked LU rounds differently in double precision, and where that flips a single-precision bit of QJH a marginal PK root lands elsewhere - 108 of the monarch print's 170,034 lines, no crossing.

The aerodynamic cache (`msc/mscaec.c`): what of a SOL 145 run depends on the structure is the modes and what is built on them (GKI, DJH, QJH, QKH, QHH, FA1); what does not is AJJ per pair (AMG) and its LU factors (AMP). With `N95_AERO_CACHE` set, `DLAMGK` in `mis/amgk.f` keeps each pair's AJJ under a 64-bit multiply-xorshift hash of every input word (the ACPT record, the counts, NJ, the symmetry flags, REFC, Mach, k, the kernel word) and `AMPKC` in `mis/ampk.f` keeps the LU factors and pivots under a hash of the AJJ; a hit goes straight to `ZGETRS`. Simcenter's PFAERO subDMAP makes the same split, and MSC and ZAERO keep the aerodynamic matrices for restarts with a changed structure.

6. The PK flutter method as FA1 does it

FA1 KHH, BHH, MHH, QHHL, CASECC, FLIST/FSAVE, KXHH, BXHH, MXHH/S, N, FLOOP/S, N, TSTART/S, N, NOCAD/V, Y, PKMATCH=0/V, Y, PKVECT=0 (`r/f/AERO10` line 566). NASA's PK is in `mis/fa1.f` (the loop list), `mis/fa1pki.f` (the interpolation tables), `mis/fa1pke.f` (the iteration), `mis/fa1pka.f` + `HSBG` + `ATEIG` (the eigenvalues) and `mis/fa1pkv.f` (the eigenvectors); `mis/fa2.f` prints. Everything in this section except where marked "fork" is NASA's arithmetic.

6.1 The flutter loops

FA1 reads the FLUTTER card the FMETHOD selects off the FLIST record (`FLUT(1..10)`: `FLUT(1)` the id, `FLUT(2)` the method, `FLUT(4)`, `FLUT(5)`, `FLUT(6)` the density, Mach and velocity `FLFACT` ids, `FLUT(7)` IMETH, `FLUT(9)` NVALUE (NEIW, the roots wanted per loop), `FLUT(10)` EPS; `mis/fa1.f` labels 30-97) and writes the loop list to `FSAVE`, three words per loop: Mach, velocity, density ratio. For PK it wrote every (density, Mach, velocity) combination, the velocity innermost, with the 100-loop stop in the DMAP (`REPT LOOPTOP, 100`, Flutter Error 3). **Fork:** for `PKNL`, or PK with `PARAM PKMATCH 1` (`PKMATC` in `/BLANK/`), one loop per entry of the density and velocity lists taken together, the Mach list walked alongside with its last entry repeated when it is shorter (`mis/fa1.f` label 98); the ninth word of the `FSAVE` header (`REC0(9) = 1`) tells FA2 the run was matched. Everything after that is NASA's PK untouched: a matched point is just a loop with its own density and velocity. For PK `SMETH = 2` always: NASA's comment reads "PK METHOD HAS LINEAR SPLINE ONLY", meaning the one-dimensional (infinite beam) spline of section 6.2, which is cubic in k, and not a piecewise-linear interpolation; the front end's UIM 9117 ("IMETH L, linear interpolation in k") is the same thing in looser words.

6.2 FA1PKI: the interpolation of Q in k

Rebuilt whenever the Mach changes (`IFLAG`). FA1PKI takes the Mach of the `QHHL` list closest to the loop's Mach, counts its NK reduced frequencies `k_i`, and builds the (NK + 1)-square matrix

$$\begin{aligned} A_{ij} &= |k_i - k_j|^3 + (k_i + k_j)^3 & (i, j \leq NK) \\ A_{i, NK+1} &= A_{NK+1, j} = 1, & A_{NK+1, NK+1} = 0 \end{aligned}$$

in double precision and inverts it (`INVERD`; a singular A is UFM 2427). The second term is the image of the kernel in $k = 0$, which makes the fit even in k. The tables hold, for each `k_j`, $\text{Re } Q(k_j)$ and $\text{Im } Q(k_j) / k_j$ - the imaginary parts are divided by their own k before interpolation (`mis/fa1pki.f` label 120). At an arbitrary k, FA1PKE forms $P_i = |k - k_i|^3 + (k + k_i)^3$, $P_{NK+1} = 1$, $C = A^{-1} P$, and

$$Q_R(k) = \sum_j \text{Re } Q(k_j) C_j, \quad Q_I(k) = \sum_j [\text{Im } Q(k_j) / k_j] C_j$$

(`mis/fa1pke.f` labels 115-145). Q_I is thus the interpolated $\text{Im } Q / k$, not $\text{Im } Q$. The same kernel, scaled by 1/12 and with a smoothing term, is MINTRP's `LSPLIN` of the gust path (section 9.3).

6.3 FA1PKE: the state matrix and the iteration

At the start of a Mach group FA1PKE unpacks `KHH`, `BHH` (zero when absent) and `MHH` into core and inverts `MHH` (`INVERS`; a singular `MHH` is `MESAGE -7`). For each loop (Mach, velocity V , density ratio) it sets

```
rho = RREF * ratio / 2      (RREF the reference density of the AERO card)
BOV = b / V,   RBV = rho b V,   RVS = rho V^2
```

and, at the trial reduced frequency k (`KINT`, starting from 0 for the first root of each loop), forms the real matrices (comments at `mis/fa1pke.f` label 145)

```
K = -KHH + (rho V^2 / 2) Q_R(k)      [the code's RVS * QR, rho already halved]
B = -BHH + (rho b V / 2) Q_I(k)      [RBV * QI, Q_I being Im Q / k]
```

and the $2N$ -square state matrix

$$A = \begin{vmatrix} 0 & I \\ M^{-1} K & M^{-1} B \end{vmatrix}$$

so that, with $K_{\text{eff}} = KHH - (\rho V^2/2) \text{Re } Q$ and $B_{\text{eff}} = BHH - (\rho b V/2) \text{Im } Q / k$, the blocks are $M^{-1} K_{\text{eff}}$ and $M^{-1} B_{\text{eff}}$ of the first-order form of $[M p^2 + B_{\text{eff}} p + K_{\text{eff}}] q = 0$. (`GMMATS` forms the products; **fork**: `FA1PKG` and `FA1PKQ` in `mis/fa1pkq.f` form the same products and eigenvalues with the same arithmetic in the same order, without the argument aliasing `FA1PKA/HSBG/ATEIG` are built on, at -O3; NASA's path is kept under `DIAG 39`.) The eigenvalues come from Hessenberg reduction (`HSBG`) and the QR algorithm (`ATEIG`): a 240-square QR per iteration per root on a 120-mode basis, 4-5 ms each, nine tenths of FA1's time. `ATEIG` reads one double outside its matrix (its search for a small subdiagonal element steps past (2,1) to `A(1-IA)`), and whether that double is below `EPS` decides where the QR sweep starts, which moves roots in the sixth digit; the fork's twin keeps the read and lays each thread's arrays out as open core is so the same bytes are there.

The $2N$ eigenvalues $p = \sigma + i \omega$ are sorted by ascending imaginary part (`RSORT`), those with negative imaginary part discarded, and the loop walks up the list:

- a root with $\omega = 0$ (a real, aperiodic root) is accepted at once, with $k = 0$, $f = 0$ and the damping $g = b \sigma / (0.34657 V)$ (label 200; the literal 0.34657 equals $\ln 2 / 2$, and its origin is not documented);
- for an oscillatory root the test value is $RKTST = BOV * \omega$, the reduced frequency the root implies. If $|RKTST - KINT| < EPS$ (`EPS` from the `FLUTTER` card, 0.001 when blank or zero) the root is accepted (label 230): the five words kept are σ , ω , $k = RKTST$, $f = \omega / 2 \pi$, $g = 2 \sigma / \omega$. Otherwise $KINT = RKTST$ and the matrices are rebuilt and the QR repeated (NASA's fixed-point iteration on k , label 220);
- after an acceptance the walk continues down the same eigenvalue list at the k the last root converged at, so the next root's first estimate comes free and is often accepted without another QR (the `N95_PK_STATS` histogram's bin 0);
- at 10 iterations without convergence (`NIT = 10`, label 240) a straight line is fitted by least squares through the (trial k , resulting k) pairs from the second trial on and its fixed point $k^* = -A10 / (A11 - 1)$ taken, with `UWM PK METHOD FIALED TO CONVERGE, n ITERATIONS ON LOOP 1, FOUND m, ROOTS WANTED n / LEAST SQUARES FIT APPROXIMATION IMPLEMENTED` (the misspelling is NASA's);
- the loop ends when `NVALUE` roots are accepted (`NROOT >= NEIW`); running out of eigenvalues first is a system fatal (`ERROR IN FA1PKE/@ 200`).

Fork, the secant step (`N95_PK_SECANT=1`): once two iterates (k_{prev} , k) with their results ($g_{\text{prev}} = k_{\text{from_root}}(k_{\text{prev}})$, $g = k_{\text{from_root}}(k)$) exist, the next trial is the secant on the residual $h(k) =$

$k_{\text{from_root}}(k) - k$, $k_{\text{next}} = k - h1 (k - k_{\text{prev}}) / (h1 - h0)$, reverting to NASA's step when the denominator is tiny, the step non-positive, or more than twice NASA's (label 225; the same arithmetic in FA1PKL). Off by default: it moves bits (section 12).

Fork, the loops side by side (mis/fa1pkp.f): since each loop is solved from $k = 0$ with nothing carried over, the loops of one Mach are independent; on the first loop of a group FA1PKP solves them all, one OpenMP thread per loop (FA1PKL, FA1PKE's labels 100 to 300 transcribed onto arrays of their own), and FA1PKE then writes loop by loop in order what the serial solve would have written (FA1PKR). N95_PK_THREADS=1, DIAG 39 and a basis of fewer than four modes keep the serial solve. The print is the serial run's to the bit, and that is how it is checked.

6.4 Matched points, damping and the quantities printed

Each accepted root is five words: Re p, Im p, k, f, g, appended per loop to BXHH (CLAMA in the PK path; mis/fa1pke.f label 310) and copied into FSAVE on the last loop. FA2 (mis/fa2.f) prints the FLUTTER SUMMARY from them, for PK reading the five words back (label 3200): KFREQ = k, $1./KFREQ$, DENSITY (the row's own density ratio on a matched-point run), MACH NO. (the row's own), VELOCITY = $|V| / VREF$ (PARAM VREF, default 1.0), DAMPING = $g = 2 \text{ sigma} / \text{omega}$, FREQUENCY = f, and the COMPLEX EIGENVALUE sigma, omega (format 1070: F9.4 then eight E13.6). **Fork:** with the ninth FSAVE word set, the summary is one block per root (POINT = n METHOD = PK (MATCHED POINTS), format 1043) with a row per matched point and the density and Mach in every row, MSC's PKNL layout; COSMIC's own PK layout is one block per (Mach, density) group. The K method's g and f are formed from the complex eigenvalue of CEAD the other way ($VOUT = |Im| / VREF$, $F = KFREQ * Im / (\pi b)$), and the KE method returns no eigenvectors (User's Manual Vol. II §3).

g is the structural damping that would have to be added to make the root neutrally stable; a flutter crossing is where a root's g crosses the threshold the plotter uses (VehicleDesign's plot_nastran_flutter; the crossing rules, drop_isolated and the two-points-above requirement are the reader's, not the solver's).

6.5 Root tracking (fork)

FA1 accepts the roots of a loop in ascending frequency, so POINT n of the summary is the n-th lowest root at that speed, and where two roots cross in frequency the columns swap: the V-g curves kink. FA1PKT (mis/fa1pkt.f) keeps every accepted root's modal eigenvector on scratch 302 - the vector of $p^2 I + p M^{-1} B + M^{-1} K$ at the root, with the matrices of the reduced frequency the root converged at, solved by EGNVCT - and FA1PKU assigns the roots of the next loop to the previous loop's places greedily over increasing cost, the cost of pairing previous root i with current root j being $(1 - MAC) + 0.25 \times \text{the relative distance of the eigenvalues}$, MAC the modal assurance criterion between the two PK eigenvectors; roots the previous loop did not have take the places left, in their own order; the previous loop's vectors travel on MXHH (data block 204). The tracking restarts on the first loop and when the Mach changes, and is skipped with a note when the work space is too small. The first version compared vectors alone and never moved anything. On the reduced monarch deck it lowers the elastic roots' mean $|\delta g|$ per step from 0.078 to 0.054 at equal frequency continuity, the same as the plotter's re-sort after the fact, and the two agree on the order. On the full deck a third to two thirds of the 57 roots change place at most loops, most at the low-speed end where the aerodynamic and rigid-body roots wander.

6.6 Eigenvectors and the aerodynamic loads

For a marked loop (a negative FLFACT velocity) FA1PKV computes, for every accepted root, the modal vector of the state matrix at the root's converged k (EIGENVECTOR FROM THE PK METHOD: the complex coefficient of each mode of the basis) and writes it to the recovery scratch 301; PARAM PKVECT 1 prints it for every loop (**fork**; the file, the print and the pass count are FA1PKW's so the threaded solve can compute in a thread

and write in order). The physical vector of any point is $\Phi_i q$ with the modes run's Φ_i . The rigid format recovers the physical complex eigenvectors of the marked loops through MODACC (matching OFREQUENCY's velocities on the imaginary part), DDR1, SDR1, MPYAD GTKA,CPHIA,/CPHIK (the boxes), UMERGE and SDR2/OFP as COMPLEX EIGENVECTOR tables; **fork:** AERO 10's restart tables give every statement from LABEL VDR to LABEL FINIS, the GO,GOD/GM,GMD EQUIV and the PFILE PARAM the flutter cards' bits (34-40), so a restart recovers the vectors like a direct run (HALO.md, "The restart tables").

ADR (ADR CPHIH1,CASEZZ,QKHL,CLAMAL1,SPLINE,SILA,USETA/PKF/BOV/C,Y,MACH = 0.0/*FLUTTER*, rf/AERO10 line 722) forms the box loads of every recovered root, $P_{kf} = Q_{kh}(k, Mach) u_h$, interpolating QKHL over k (through ADRI, MINTRP and the spline of section 9.3) at the root's reduced frequency $k = \omega BOV$ and at the Mach of the MKAERO1 list closest to PARAM MACH (NASA's default 0.0 took the lowest Mach for every subcase; the driver writes each child's own). ADRPRT prints them as AERODYNAMIC LOADS (UNIT DYNAMIC PRESSURE): per box the real and imaginary parts of T1..T3 then R1..R3, per unit dynamic pressure, in the box's frame. BOV comes from the AERO card's velocity, one number per run, which is why one marked loop per subcase gives every recovered root its own k and several marked loops give only the first loop's loads at the right k (UWM 9456). On the two-subcase test deck the beam's in-plane modes show 1e-17 and the out-of-plane ones order one to sixty, which is the physics.

7. What a SOL 145 child costs, and where the fork's time goes

One child (one Mach) runs the setup (GP1-GP4, MCE1/MCE2 with a DECOMP, READ, GKAD/GKAM, APD, GI; about 6 s serial at -Oo), AMG, AMP, FA1, FA2 and, for marked loops, the physical recovery (OFP formatting of a 1.7-million-line print took 23 s). On the five-Mach monarch deck at 40.5 s wall the CPU is FA1 about 650 s of 956 (nearly all NASA's QR), AMP 170, AMG 100; FA1 is bound by its own dependences and the order of its operations is the answer - a faster QR (LAPACK's DHSEQR was tried: slower at $n = 220$ and 7 % of eigenvalues differing in the last single-precision bit) would not be NASA's roots. Fewer QR solves per root is the remaining lever (N95_PK_SECANT), and fewer roots (NVALUE) at deck level (skill nastran95-performance §2, §6, §9).

8. The checkpoint and the restart

NASTRAN's restart tables re-execute a DMAP statement on a modified restart when a card of one of the statement's ****CARD bits changed (the table under \$*CARD BITS in rf/AERO10: bit 36 is FLFACT/FLUTTER, 34-40 the flutter cards together; bit 25 MACH, 54 GUSTAERO, 62 HFREQ LFREQ LMODES KDAMP). A rigid-format-switch restart (RF3 -> AERO10, UIM 4145) off a scr=no modes run skips GP1 through READ and runs GKAM, APD, GI, AMG, AMP, FA1, FA2 when the restart deck's own cards are aero and flutter cards only; any other new card is a change to the structure and GP1 runs again. The modal basis off the tape and the one a direct run's READ computes in the same job are FEER's answer to the last bit, not to the last printed digit, so the restarted summaries agree with a direct run's to six significant figures on the monarch (258 of 1,830 rows of a 30-mode deck differ in the seventh digit) and digit for digit on the two-subcase test deck. An ALTERed OUTPUT4 has no restart-table entry and is not executed on a restart, so the export needs a cold run (HALO.md, "The modes once"; "Known (2026-10-01)").

9. The gust response (AERO 11)

9.1 The chain

RF3 front end -> DPD (the dynamic loads table DLT, PSD list PSDL, frequency list FRL, transient list TRL) -> READ -> GKAM -> APD -> GI -> AMG -> AMP (QHHL, QKHL, and QHJL only with GUSTAERO = +1) -> FRLG (the direct loads in modal coordinates per frequency; transient loads Fourier-transformed) -> GUST ("forms the loading due to gusts and adds to the direct loads": inputs DLT, FRL, DIT, QHJL, ACPT, CSTMA, BOV, MACH, Q) -> FRRD2 solving

$$[-M_{hh} \omega^2 + i B_{hh} \omega + K_{hh} + q Q_{hh}(k)] u_h = P_h(\omega)$$

-> IFT (the inverse Fourier transform for a transient formulation; PARAM IFTM 0 rectangular, 1 trapezoidal, 2 cubic spline) -> MODACC -> ADR (frequency only) -> the recovery -> XYTRAN/XYPLOT -> RANDOM (PSD and autocorrelation from the RANDPS cards and the XY requests; the RMS, the expected zero-crossing frequency No) (User's Manual Vol. II §4, §5). PARAM Q (dynamic pressure, required) and PARAM MACH are C,Y constants: one each per execution. The gust aerodynamics are computed for GUSTAERO = +1 in NASTRAN-95 and -1 in MSC, so the front end flips the sign. What the GUST card defines (profile, amplitude, reference velocity, lag) and whether a spanwise-varying gust exists are in Vol. I, not on disk: not documented. The repo's PSD decks carry the von Karman spectrum as MSC's TABRNDG, which the front end passes through unchanged; the 1-cos and turbulence decks are TLOAD / TABLED1 time histories through the Fourier/IFT pair.

9.2 The gust solve in core (fork)

AMPF solved $R_{JH} = A_{JJ}^{-1} S(k)$ per pair through CFACTR + CFB5OR after the same k-squared file walks as AMPC; it now takes the in-core solve without the transpose (AMPCZN in mis/ampcz.f) and keeps its file positions. NASA's AERO 11 demos d11031a and d11032a agree with the out-of-core path to 1.4e-4 and 1e-5 of the largest value on each line.

9.3 The spline in k, and why it is solved in double precision (fork)

ADRI interpolates the gust matrix QHJ (and ADR's QKH) over reduced frequency, FRD2I interpolates QHH, both through MINTRP -> LSPLIN: for target k_d the weights of the NI tabulated k_i are the first NI components of $A^{-1} R(d)$, with

$$A = \begin{bmatrix} K(y_i, y_j) + DZ \delta_{ij} & 1 \\ 1 & 0 \end{bmatrix} \quad R(d) = \begin{bmatrix} K(y_d, y_i) \\ 1 \end{bmatrix}$$

$$K(a, b) = (|a - b|^3 + |a + b|^3) / 12$$

- the kernel and its image in $k = 0$, so the fit is even in k ; $KY = 1$, $KD = 0$, $KT = 1$, no slope terms, one constant term (mis/lsp1nd.f header). LSPLIN solved the system by INVERS in single precision, and with k values close together (the monarch's list starts 0.005, 0.01, 0.02, and every value sits next to its image) the system is ill-conditioned: the weights came out of order 1-20, oscillating, and 0.988 at a tabulated k that is a node. Rows of QHJ whose imaginary part bends at low k (boxes far aft) came out up to 70 % off at the tabulated k themselves, the gust response was jagged from frequency to frequency above 1.5 Hz, and against Simcenter 2606 accelerations and forces were 3-24 % off in rms and No up to 48 % off, while the point-force response through the smoother QHH was within 0.3 %. LSPLND solves the same system (the same kernel, constant term and weights) by Gaussian elimination with partial pivoting in double precision; ADRI and FRD2I switch it on (MINTDQ) and FA1's K method keeps

LSPLIN. An MKAERO list holding every analysis k exactly now gives the interpolated run's answers (HALO.md, "SOL 146 random response").

9.4 The column that stopped at row 65,536 (fork)

UNPACK's first-to-last mode (the row range left to the column) ended at NASA's LARGE = 65536, silently closing a column there, and MPYAD method 10 (MMA1) reads every column of "A" that way. ADRI's interpolation hands MINTRP QHJL as one column per k of boxes x modes rows (2,282 x 57 = 130,074 on the monarch), so the interpolated QHJK summed to 2.5e-6 where its input summed to 7,174 and the gust loads were zero; a 1-cos run answered only its dummy DAREA. The same limit zeroed ADR's AERODYNAMIC LOADS (QKHL, 4,564 x 57 rows). LARGE is now 2^30 (mds/unpack.f); any product with "A" taller than 65,536 rows (a model past about 10,900 grids, or a flattened matrix of that height) was affected on an older executable (HALO.md, "Tall columns").

10. OUTPUT4: the matrices exported and their conventions

msc/mscop4.c writes the alter after SDR1 (statement 77 of rigid format 3, where PHIG and MGG both exist) or after AMP (statement 90 of AERO 10 and AERO 11, where QHHL, QHJL, QKHL, MHH, KHH, BHH and PHIDH exist), as OUTPUT4 <block>,,,//-1/<unit>/2, and rewrites NASTRAN-95's formatted file into MSC's layout.

block	rows x columns	meaning
PHIG (PHG)	g-set dof x modes (cut to the modes requested)	the real eigenvectors, MASS-normalised
MGG, KGG	g-set x g-set	the mass and stiffness, WTMASS applied; null columns left out as MSC does (the monarch_ff .phg has 150,166 non-zeros to MSC's 150,240, all exact zeros in the rigid-body columns; the 5,777 mass non-zeros match exactly)
PHIA, MAA, KAA	a-set	the same on the analysis set
QHHL (QHH)	n_h x (n_h x n_k) per Mach	the generalized aerodynamic force matrices of every (Mach, k) pair of the child's list, blocks side by side in the sorted order of UIM 9457, per unit dynamic pressure, complex (type 4, real/imaginary pairs)
QKHL (QKH)	k-set x (n_h x n_k)	the box forces per mode per pair
QHJL (QHJ)	gust columns per pair	the gust aerodynamics (with GUSTAERO = +1)
MHH, KHH, BHH	n_h x n_h	the modal mass, stiffness and damping of GKAM
PHIDH (PHDH)	d-set x n_h	the modal matrix on the d-set

Conventions and checks:

- Q multiplies the modal displacements to give the modal aerodynamic force per unit dynamic pressure; $k = \omega c / 2V$, $b = REFC / 2$. The aerodynamic stiffness enters the flutter equation as $K_{eff} = KHH - (\rho V^2 / 2) Re Q$, the aerodynamic damping as $B_{eff} = BHH - (\rho b V / 2) Im Q / k$ (section 6.3).
- The export was checked by replaying FA1's own arithmetic on it: the cubic spline in k through $Re Q$ and $Im Q / k$, the two matrices above, at each printed root's own reduced frequency, gives the printed

PK roots back to $1e-4$ (VehicleDesign's `test_ase_plant`).

- Simcenter's export of the same deck agrees with the fork's to 5.5 % of the matrix at low k (a steady offset of the plunge mode's small real part) and under 1 % at $k = 3.2$, once the modes' signs are matched (each child solves its own modes; `modal_sign_match`).
- A Simcenter deck's alter inside the AMP loop appends `QHH` once per pair, every subcase over again; the fork ignores the alter text and maps `QHH / QHJ / QKH` to the whole lists, one file per subcase (`<name>_s<subcase>.op4`), each child's `QHHL` holding its own Mach's blocks in that Mach's `MKAERO1` order after APD's sort.
- The file: header `NCOL NROW FORM TYPE (4I8), name (A8), 1P,5E16.9`; `FORM 1` square, `2` rectangular, `6` symmetric (NASTRAN-95's negative form made positive); real out as type `2`, complex as type `4`; per column `IC IR NW (3I8)` then `NW` values five to a line; the trailer column `NCOL+1 1 1` and one `1.0`. The three record traps in NASTRAN-95's own file, all from `mis/oupt4.f`: records are fixed-width (`1X,3I13 / 1X,10E13.6` single, `1X,3I16 / 1X,8D16.9` double) and must be sliced at the field width because adjacent negatives touch; `JJ` counts single-precision words, so a double column announces twice the values; a null column comes back with `II` zero and the previous column's words in the buffer and is read past and left out.
- Restart limitation: the alter is not executed on a modified restart (section 8).

11. Sign conventions summarised

quantity	convention	source
k	$\omega b / V$, $b = REFC / 2$	<code>mis/fa1pke.f</code> (<code>BOV = BREF/VEL</code>), <code>skill nastran-flutter §8</code>
g (oscillatory root)	$2 \sigma / \omega$	<code>mis/fa1pke.f</code> label 230
g (real root)	$b \sigma / (0.34657 V)$	<code>mis/fa1pke.f</code> label 200
f	$\omega / 2 \pi$	<code>mis/fa1pke.f</code>
VELOCITY printed	$V / PARAM VREF$	<code>mis/fa2.f</code>
Q	force per unit dynamic pressure on the modal displacements; <code>Im Q / k</code> is what FA1 tabulates and interpolates	<code>mis/fa1pki.f</code>
GUSTAERO	+1 computes the gust aerodynamics in NASTRAN-95, -1 in MSC/Simcenter; the front end flips it	<code>msc/mscxlat.c</code> UIM 9119
KDAMP	-1 (default) viscous <code>b_i</code> ; +1 complex stiffness in NASTRAN-95 - the opposite of MSC's numbering; passed unchanged	User's Manual Vol. II §3, fork <code>GKAM1A</code>
CPHIK rows	plunge in T_3 , pitch in R_2 , in the box's frame	<code>HALO.md</code>
AEROF	real and imaginary $T_1..T_3$, $R_1..R_3$ per box per unit dynamic pressure, at the root's k	<code>HALO.md</code> , <code>ADRPRT</code>
ids above $2^{24} - 1$	renumbered in the deck, restored in the print and the <code>OUTPUT4</code> names	<code>msc/mscxlat.c</code> , <code>msc/mscf06.c</code>

12. The verification record

Every number here is quoted from the sources named; "line for line" means `compare_prints.sh`'s 0 differing lines with only clock and date lines excluded.

12.1 The solver and the front end against NASA and MSC

check	result	source
NASA's 132 demonstration decks against NASA's 1995 prints	70 reproduce NASA's output after the RFDIR truncation fix (55 before); the four-way test's verdicts: 85 match, 19 nothing to compare, 18 fatal, 10 differ, none unstable since the libgfortran format-copy wrapper	HALO.md ; skill build-nastran95
the four fluid demos (d03021a, d03031a, d07021a, d07022a)	48 of 48 runs at twelve environment sizes complete and match NASA with the <code>--wrap</code> of <code>fc_strdup_notrim</code> ; 4 of 36 failed without it	HALO.md
Linux against Windows	every number of the licence-free test the same, except demo d03031a where FEER returns 30 of NASA's 35 roots on Linux (a convergence decision on the last bit, deterministic)	skill build-nastran95
the weight generator against MSC Nastran 2025.1	digit for digit, including <code>PARAM,GRDPNT</code> after renumbering (a missed renumbering had put the two c.g.s 0.15 m apart)	HALO.md
the repo's modal decks against MSC	flexible modes paired at $MAC > 0.99$ with frequencies within $1e-4$ relative; <code>phi ' M phi</code> the identity through the exported <code>.phg / .mgg</code>	skill nastran-analysis §6; HALO.md
SOL 200 on MSC's three-bar truss	MSC's design-cycle history within half a percent at every cycle	HALO.md

12.2 Flutter

check	result	source
NASA's doublet-lattice demos d10021a-d10023a (KE and PK, 15-degree swept wing), in-core against out-of-core solve	every flutter summary number to $1e-5$ relative or better (single against double precision); the optimised kernel bit for bit with <code>-Oo</code>	HALO.md
the reduced monarch deck (Mach 0.10, 31 k, 30 modes solved, 20 kept)	the 30 modes agree with MSC's to every printed digit; the first flexible root's frequency and damping to three digits at every sampled point (the first <code>flutter_vs_msc</code> run)	HALO.md ; skill nastran-flutter §7
K2PP / B2PP steady-lift DMIGs (the trim-load study, 8 subcases x 3 points) against Simcenter 2606	the outer-boom pair agrees at every airspeed to the printed digits; the crossing 18.037 m/s at 4.185 Hz in both; the zero-load twin gives the no-load numbers	HALO.md
the five-Mach <code>monarch_demo_asm1083</code> deck against Simcenter 2606 (benchmark records)	the lowest crossings agree within 0.1 % on the same (quartic) kernel (the VehicleDesign repo guide, 2026-09-30: crossing speeds within 0.1 %, frequencies within 0.02 %); the quartic kernel moves Mach 0.40's lowest crossing from 16.17 m/s at 38.4 Hz (a PK artefact) to 19.77 m/s at 10.70 Hz, Simcenter's root	skill nastran-flutter §7; skill build-nastran95
the same deck through ZAERO's g-method on the ZAERO chain's deck	the same mechanisms 1-4 % lower in EAS	VehicleDesign repo guide (CLAUDE.md)
HALE wing (Patil, Hodges, Cesnik): flutter, frequency, divergence	quartic 31.5679 m/s, 3.7329 Hz, 39.8809 m/s against Simcenter 31.5679, 3.7329, 39.8810; parabolic (both) 30.6187 / 30.6188, 3.8225, 39.8805 / 39.8806	HALO.md

check	result	source
Goland wing	155.2405 m/s, 10.9352 Hz against Simcenter 155.2401, 10.9352; parabolic 154.4496 / 154.4491, 10.9729	HALO.md
TKERQ against PanelAero's kernel, 3,000 random geometries	1.6e-7 (P1), 9.5e-6 (P2); INCROQ against a Python element 1.1e-4 (single-precision output)	HALO.md
the lowest crossings per Mach held by test_nastran95ase_aeroelastic (EAS m/s / Hz, Mach 0.10 .. 0.40)	parabolic 33.326/3.721, 29.900/3.812, 22.331/10.738, 20.116/10.782, 19.754/10.735; quartic 33.306/3.710, 29.844/3.799, 22.410/10.699, 20.152/10.741, 19.784/10.699 (0.06-0.35 % apart)	skill nastran95- performance §1
NVALUE 83 of 110 against all 110 on the 3 x 3 study	the roots found in 0.6-20 Hz the same bits: 1,836 of 1,836 rows identical	skill nastran95- performance §4

12.3 The speed-ups, the cache, the kernels, the restart

check	result	source
every speed-up but the OpenBLAS LU, the five-Mach deck (170,034 lines)	identical to the unmodified source; with N95_AJJ_SOLVE=builtin identical to Jon's halo-ase- sol1145 8cd363e line for line	HALO.md, "Linux"
the OpenBLAS LU	108 of 170,034 lines (marginal PK roots); none of the 35 crossings (29 with drop_isolated); on the 110-mode PKVECT deck 48 of 22,447 summary rows plus eigenvector last digits, every crossing the same	HALO.md; skill nastran95- performance
the aerodynamic cache, filling and hitting, against the cache off	0 differing lines on the 110-mode PKVECT deck (10,070,885 lines) and the study deck (154,119 lines)	HALO.md
the quartic path off (N95_DLM_QUARTIC=0) against the build before it	0 of 10,070,885 lines; NASA's 132 demos as before	HALO.md
the secant off against the build without it	identical, the demos too; on, serial and threaded bit-identical; the lowest crossing moves at most 0.001 m/s	HALO.md
the UNPACK fix	the five-Mach print identical outside the AERODYNAMIC LOADS pages (6,764,231 lines); the demos as before	HALO.md
the double-precision spline	of the 132 demos only d11031a and d11032a move, below 1.4e- 4 of a line's largest number	HALO.md
the restart flow against the direct run	six significant figures on the monarch; digit for digit on the two-subcase deck; the speed-ups reproduce each flow exactly	HALO.md
Windows parity build against Jon's exe (63e15d64), the 110- mode deck	N95_AJJ_SOLVE=builtin N95_DLM_QUARTIC=0: 0 of 10,035,296 lines in 673.7 s against 1768.6 s; OpenBLAS parabolic 8 lines at 1e-21 in 305.8 s, the same crossings	skill build- nastran95
the unmodified solver's own nondeterminism	do1002a prints the GINO timing constants it measures; do7021a/do7022a take 10 complex decompositions instead of 13 in 1-3 runs of 40 on every build; UIM 3028's decomposition statistics vary	skill nastran95- performance §5

12.4 Gust

check	result	source
von Karman PSD deck monarch_demo_asm1083_gust_h0km_eas8p5 against Simcenter 2606	every curve's rms within 0.3 % (the root bar's item 3 2.6 %; Simcenter adds its default residual vector, the fork none), No within 3.5 %; the tip transfer function within 0.3 % to 2.7 Hz and within 9 % and 10 degrees at the 4.3 Hz mode pair	HALO.md ; skill nastran-flutter §7
1-cos deck ..._gust1cos_h0km_eas8p5	tip extremes above grid 1010001 within 0.05 % (-94.19 against -94.17 mm; 0.8-1.7 % before the spline fix, -2.3 mm before the UNPACK fix); plunge -8.120..8.071 m against -8.120..8.070 m; correlation 0.9996 over the 6 s before the spline fix	HALO.md ; skill nastran-flutter §7
turbulence deck ..._gustvk_h0km_eas8p5_s1	tip rms 29.652 against 29.663 mm, extremes within 0.4 %, correlation 0.99999, plunge within 0.7 mm	skill nastran-flutter §7
the gust in-core solve, NASA's d11031a / d11032a	1.4e-4 / 1e-5 of each line's largest value against the out-of-core path; d11031a nearer NASA's 1995 print than before	HALO.md

13. Not implemented, and open

- **The g-method** (ZAERO's; Chen, AIAA J. 38(9), 2000): HALO.md sets out what it takes in FA1's PK branch - $Q'(ik)$ per pair by central differences of the interpolated Q_{HH} , the quadratic-in-g eigenproblem linearised and swept in k through CEAD, FA2 summaries per root of the sweep, a sixth FLUTTER method name - and records it as not started.
- **SOL 144** (static aeroelastic trim): no COSMIC counterpart.
- **CAERO4** (strip theory): not in the front end.
- **KE and K methods**: NASA's, untouched; PKMATCH applies to PK only.
- **Residual vectors**: NASTRAN-95 never had them; Simcenter's default residual vector is the stated reason for the 2.6 % on the root bar's PSD item.
- **AMG's drivers at -O2**: the loops over boxes in amgb1a / amgb1 and the GINO packing of a 2.6 GB AJJL stay at -Oo (an -O2 driver produced an unusable AJJ); the whole tree at -O2 prints 124 of 132 demos differently and segfaults on 51, because open core is overlapping dummy arguments.
- **Blocked ATEIG, bigger GINO buffers, thread priorities, x86-64-v4, more AMP slots**: tried and dropped (skill nastran95-performance §6).

14. Where to read further

topic	file
the fork's engineering log, every change with its reason and check	HALO.md
the PK iteration, the loop list, the summary	mis/fa1pke.f, mis/fa1.f, mis/fa2.f, mis/fa1pki.f

topic	file
the fork's PK additions: threads, the QR twin, tracking, vectors	<code>mis/fa1pkp.f</code> , <code>mis/fa1pkq.f</code> , <code>mis/fa1pkt.f</code> , <code>mis/fa1pkv.f</code>
the doublet lattice and its batched, threaded, quartic forms	<code>mis/gendp.f</code> , <code>mis/amgk.f</code> , <code>mis/tkerv.f</code> , <code>mis/dlmq.f</code> , <code>mis/incro.f</code>
the generalized aerodynamics and the in-core solves	<code>mis/ampk.f</code> , <code>mis/ampcz.f</code> , <code>mis/ampczs/</code> , <code>mis/ampf.f</code>
the aerodynamic cache	<code>msc/mscaec.c</code>
FEER's guards	<code>mis/ferxtd.f</code> , <code>mis/ferxts.f</code> , <code>mis/fqrvv.f</code> , <code>mis/fqrv.f</code> , <code>mis/fernpd.f</code>
the gust spline and the tall-column fix	<code>mis/lsplnd.f</code> , <code>mis/mintrp.f</code> , <code>mis/adri.f</code> , <code>mis/frd2i.f</code> , <code>mds/unpack.f</code>
the (Mach, k) sort	<code>mis/apd.f</code> (APDMKS)
the rigid formats, their parameters and card bits	<code>rf/AERO10</code> , <code>rf/AERO11</code> , <code>rf/DISP03</code>
the translation, the driver, the OUTPUT4 rewrite, the print rewrite	<code>msc/mscxlac.c</code> , <code>msc/mscexec.c</code> , <code>msc/mscflut.c</code> , <code>msc/mscop4.c</code> , <code>msc/mscf06.c</code>
NASA's own account of the formats	NASTRAN Users Manual 2.pdf (Vol. II, SP-222(08)), distilled in VehicleDesign's <code>nastran-analysis</code> skill; the Theoretical and Programmer's Manuals are not distilled